

Lecture 07 — The Heston Model: CIR Variance, Affine Structure, and Simulation

Computational Finance (WI4151), Cluster 4 — Models

These notes cover the Heston stochastic volatility model: the SDE system, the CIR variance process and the Feller condition, the affine machinery that produces the characteristic function, and the simulation schemes (Euler with truncation/reflection, exact Broadie–Kaya). The single most important object for this course is the *Heston characteristic function* of $\ln S_T$, because it is exactly the input the COS method needs to price under Heston. Cross-reference: [notes/Lecture06-notes.pdf](#) (the COS pricing formula and the cumulant-based truncation range $[a, b]$).

1 Why stochastic volatility

Black–Scholes assumes a constant volatility σ . Empirically, option-implied volatility depends on strike and maturity (the volatility *smile/skew*), so a single σ cannot reprice the whole surface. Two standard cures: add jumps to the geometric Brownian motion (Lecture 8), or make the volatility itself stochastic. The Heston model takes the second route. Its appeal over competitors such as SABR ($dS_t = \sigma_t S_t^\beta dW_t$, $d\sigma_t = \alpha \sigma_t dZ_t$) is that it admits a *semi-analytic* pricing solution (through its characteristic function) and an *exact* simulation scheme. SABR relies on the Hagan asymptotic formula, which is only accurate for a limited range of maturities.

2 The Heston SDE system

Under the risk-neutral measure $\mathbb{Q}$ the Heston model is the two-dimensional diffusion

$$dS_t = rS_t dt + \sqrt{\nu_t} S_t dW_t^S, \quad d\nu_t = \kappa(\bar{\nu} - \nu_t) dt + \gamma\sqrt{\nu_t} dW_t^\nu, \quad (1)$$

with the two Brownian motions *correlated*,

$$dW_t^S dW_t^\nu = \rho_{S,\nu} dt. \quad (2)$$

Here ν_t is the instantaneous *variance* (not the volatility; the volatility is $\sqrt{\nu_t}$). The parameters are:

- $\bar{\nu}$ — the long-run mean of the variance;
- $\kappa > 0$ — the speed of mean reversion of ν_t to $\bar{\nu}$;
- $\gamma > 0$ — the volatility of variance (“vol-of-vol”), which governs the curvature of the smile;
- $\rho_{S,\nu}$ — the spot/variance correlation, which governs the *slope* (skew) of the implied-volatility curve. As $\rho_{S,\nu}$ becomes more negative the skew steepens; this is the leverage effect.

The variance follows a Cox–Ingersoll–Ross (CIR) process: mean-reverting with a diffusion coefficient $\gamma\sqrt{\nu_t}$ that vanishes as $\nu_t \rightarrow 0$, which is what keeps the process non-negative.

Log-spot form via Cholesky. For pricing and simulation it is convenient to (i) move to $X_t = \ln S_t$ and (ii) decorrelate the Brownian drivers. By Itô's lemma on $X_t = \ln S_t$, with $dS_t/S_t = r dt + \sqrt{\nu_t} dW_t^S$ and $(dS_t/S_t)^2 = \nu_t dt$,

$$d \ln S_t = \left(r - \frac{1}{2}\nu_t\right) dt + \sqrt{\nu_t} dW_t^S. \quad (3)$$

Now write the correlated pair (W^S, W^ν) in terms of two *independent* Brownian motions $(\widetilde{W}_1, \widetilde{W}_2)$. For the 2×2 correlation matrix

$$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} = LL^\top, \quad L = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{pmatrix},$$

($\rho \equiv \rho_{S,\nu}$) the Cholesky construction $W = L\widetilde{W}$ gives $W_t^S = \widetilde{W}_1(t)$ and $W_t^\nu = \rho \widetilde{W}_1(t) + \sqrt{1-\rho^2} \widetilde{W}_2(t)$. One checks $\text{Cov}(W_t^S, W_t^\nu) = \rho \mathbb{E}[(\widetilde{W}_1)^2] = \rho t$, hence correlation ρ as required. Inverting so that the *variance* is driven by its own clean increment W^ν , the equivalent system is

$$d \ln S_t = \left(r - \frac{1}{2}\nu_t\right) dt + \sqrt{\nu_t} \left(\rho_{S,\nu} dW_t^S + \sqrt{1-\rho_{S,\nu}^2} dW_t^\nu\right), \quad (4)$$

$$d\nu_t = \kappa(\bar{\nu} - \nu_t) dt + \gamma\sqrt{\nu_t} dW_t^\nu, \quad (5)$$

where now W^S and W^ν on the right are *independent*. This is the form used for both Euler simulation and the affine derivation.

3 The CIR variance process

3.1 Conditional distribution

A central fact: conditional on ν_t , the future variance ν_T has a known distribution. Writing $\tau = T - t$,

$$\nu_T \mid \nu_t \sim \frac{Y}{2c}, \quad c = \frac{2\kappa}{\gamma^2(1 - e^{-\kappa\tau})}, \quad (6)$$

where Y follows a *noncentral chi-squared* distribution $\chi_d^*(\lambda)$ with

$$d = \frac{4\kappa\bar{\nu}}{\gamma^2} \text{ degrees of freedom,} \quad \lambda = 2c e^{-\kappa\tau} \nu_t \text{ noncentrality.} \quad (7)$$

Recall $\chi_d^*(\lambda)$ is the law of $\sum_{i=1}^d X_i^2$ where the X_i are independent $N(\mu_i, 1)$ and $\lambda = \sum_i \mu_i^2$. Its density is

$$f(x; d, \lambda) = \frac{1}{2} e^{-(x+\lambda)/2} \left(\frac{x}{\lambda}\right)^{d/4-1/2} I_{d/2-1}(\sqrt{\lambda x}), \quad (8)$$

with I_ν the modified Bessel function of the first kind, $I_\nu(y) = (y/2)^\nu \sum_{j \geq 0} \frac{(y^2/4)^j}{j! \Gamma(\nu+j+1)}$.

3.2 The Feller condition

Remark 1 (Feller condition). *If*

$$2\kappa\bar{\nu} \geq \gamma^2 \quad (9)$$

then ν_t stays strictly positive almost surely (the origin is inaccessible). If it is violated, ν_t can reach 0 (and, for the pure CIR process, instantaneously reflects / leaves 0 again; in either case ν_t never goes negative).

The cleanest way to see why this is the relevant threshold is through the density (8). As $x \rightarrow 0^+$, $I_\nu(\sqrt{\lambda x}) \sim (\sqrt{\lambda x}/2)^\nu / \Gamma(\nu + 1)$, so the density behaves like $x^{d/4-1/2} \cdot x^{(d/2-1)/2} = x^{d/2-1}$ up to constants (combining the explicit power and the Bessel power). For the density not to blow up at the origin we need the exponent $\frac{d}{2} - 1 \geq 0$, i.e.

$$q_F := \frac{2\kappa\bar{\nu}}{\gamma^2} - 1 \geq 0 \iff 2\kappa\bar{\nu} \geq \gamma^2, \quad (10)$$

recovering the Feller condition. Interpretation: a large vol-of-vol γ relative to the restoring force $\kappa\bar{\nu}$ makes excursions to zero likely; a strong restoring force keeps the variance away from the origin. This matters in practice because the non-central chi-squared exact scheme is only guaranteed well-defined when the Feller condition holds (it is a *sufficient* condition; rare exceptions exist but are not of practical interest).

4 Affine structure and the characteristic function

4.1 Affine diffusions: the general result

An n -dimensional diffusion $dX_t = \mu(X_t)dt + \sigma(X_t)dW_t$ is *affine* if the drift and the diffusion covariance are affine in the state:

$$\mu(x) = K_0 + K_1x, \quad K_0 \in \mathbb{R}^n, \quad K_1 \in \mathbb{R}^{n \times n}, \quad (11)$$

$$(\sigma(x)\sigma(x)^\top)_{ij} = (H_0)_{ij} + (H_1)_{ij} \cdot x, \quad H_0 \in \mathbb{R}^{n \times n}, \quad H_1 \in \mathbb{R}^{n \times n \times n}, \quad (12)$$

and the short rate is affine, $R(x) = \rho_0 + \rho_1 \cdot x$. The key theorem (Duffie–Pan–Singleton 2000) states that the discounted transform

$$\psi^X(u, x, t, T) = \mathbb{E} \left[\exp \left(- \int_t^T R(X_s) ds \right) e^{u \cdot X_T} \mid \mathcal{F}_t \right] = e^{\alpha(u, t, T) + \beta(u, t, T) \cdot x}, \quad (13)$$

i.e. the transform is *exponential-affine* in the state. The coefficients α, β solve a system of complex ODEs (the Riccati system). For a pure diffusion (no jumps) the system is

$$\dot{\beta} = \rho_1 - K_1^\top \beta - \frac{1}{2} \beta^\top H_1 \beta, \quad (14)$$

$$\dot{\alpha} = \rho_0 - K_0 \cdot \beta - \frac{1}{2} \beta^\top H_0 \beta, \quad (15)$$

with boundary conditions $\beta(u, T, T) = u$ and $\alpha(u, T, T) = 0$. Here $\beta^\top H_1 \beta$ is the n -vector with k -th component $\sum_{ij} \beta_i (H_1)_{ijk} \beta_j$.

Where the ODEs come from. The Feynman–Kac theorem says ψ^X (as a function of (t, x)) satisfies the backward PDE

$$\partial_t \psi + \mathcal{L} \psi - R(x) \psi = 0, \quad \psi(T, x) = e^{u \cdot x},$$

where $\mathcal{L}$ is the generator $\mathcal{L}g = \sum_i \mu_i \partial_{x_i} g + \frac{1}{2} \sum_{ij} (\sigma \sigma^\top)_{ij} \partial_{x_i x_j} g$. Substituting the ansatz $\psi = e^{\alpha + \beta \cdot x}$ gives $\partial_{x_i} \psi = \beta_i \psi$, $\partial_{x_i x_j} \psi = \beta_i \beta_j \psi$, and $\partial_t \psi = (\dot{\alpha} + \dot{\beta} \cdot x) \psi$. Dividing by ψ and collecting the terms that are constant in x versus linear in x (using that $\mu, \sigma \sigma^\top$ and R are affine) decouples into the two ODE families (14)–(15): the linear-in- x part gives $\dot{\beta}$, the constant part gives $\dot{\alpha}$. The affine structure is exactly what makes the x -dependence separate cleanly.

4.2 Is Heston affine? Restoring it in the log-domain

In the (S, ν) coordinates Heston is *not* affine: the diffusion covariance is

$$\sigma\sigma^\top = \begin{pmatrix} \nu_t S_t^2 & \gamma \rho_{S,\nu} \nu_t S_t \\ \gamma \rho_{S,\nu} \nu_t S_t & \gamma^2 \nu_t \end{pmatrix},$$

whose top-left entry $\nu_t S_t^2$ is not affine in (S_t, ν_t) . The fix is to switch to $X_t = \ln S_t$. With state $(\ln S_t, \nu_t)^\top$ the drift becomes

$$\begin{pmatrix} d \ln S_t \\ d\nu_t \end{pmatrix} = \left(\underbrace{\begin{pmatrix} r \\ \kappa \bar{\nu} \end{pmatrix}}_{K_0} + \underbrace{\begin{pmatrix} 0 & -1/2 \\ 0 & -\kappa \end{pmatrix}}_{K_1} \begin{pmatrix} \ln S_t \\ \nu_t \end{pmatrix} \right) dt + \sqrt{\nu_t} L d\widetilde{W}_t,$$

which is affine in the drift, and the covariance becomes

$$\sigma\sigma^\top = \nu_t \begin{pmatrix} 1 & \gamma \rho_{S,\nu} \\ \gamma \rho_{S,\nu} & \gamma^2 \end{pmatrix},$$

affine in ν_t . Matching the template: $H_0 = 0$ (the 2×2 zero matrix), and H_1 is the $2 \times 2 \times 2$ tensor whose first slice (coefficient of $\ln S$) is the zero matrix and whose second slice (coefficient of ν) is $\begin{pmatrix} 1 & \gamma \rho_{S,\nu} \\ \gamma \rho_{S,\nu} & \gamma^2 \end{pmatrix}$. So in log-coordinates Heston *is* affine, and the general transform result applies.

4.3 The variance ODE solved explicitly (warm-up)

Before the full Heston cf, it is instructive to solve the one-dimensional CIR Riccati exactly, which is what underlies the variance distribution. For the cf of ν_T alone, $\psi^\nu(u, \nu_t, t, T) = \mathbb{E}[e^{u\nu_T} | \nu_t] = e^{\alpha_1(t) + \beta_1(t)\nu_t}$, with $K_0 = \kappa \bar{\nu}$, $K_1 = -\kappa$, $H_0 = 0$, $H_1 = \gamma^2$, $\rho_0 = \rho_1 = 0$. Using the reparametrization $\psi^\nu = e^{\alpha(\tau) + \beta(\tau)\nu_t}$ in $\tau = T - t$, the Riccati equation reads

$$\dot{\beta} = -\kappa\beta + \frac{1}{2}\gamma^2\beta^2, \quad \beta(0) = u. \quad (16)$$

This is solvable by the substitution $g = 1/\beta$, which linearizes it: $\dot{g} = -\dot{\beta}/\beta^2 = \kappa g - \frac{1}{2}\gamma^2$. The integrating factor $e^{-\kappa t}$ gives $d(ge^{-\kappa t}) = -\frac{1}{2}\gamma^2 e^{-\kappa t} dt$, hence

$$g(t)e^{-\kappa t} - g(0) = -\frac{\gamma^2}{2} \int_0^t e^{-\kappa s} ds, \quad g(t) = e^{\kappa t} \left(g(0) - \frac{\gamma^2(1-e^{-\kappa t})}{2\kappa} \right),$$

with $g(0) = 1/u$. Inverting,

$$\beta(t) = \frac{ue^{-\kappa t}}{1 - \frac{u\gamma^2}{2}\xi_\kappa(t)}, \quad \xi_\kappa(t) = \frac{1 - e^{-\kappa t}}{\kappa}. \quad (17)$$

Then $\dot{\alpha} = \kappa \bar{\nu} \beta$ integrates (noting $e^{-\kappa s} = \xi'_\kappa(s)$) to

$$\alpha(t) = \kappa \bar{\nu} u \int_0^t \frac{e^{-\kappa s}}{1 - \frac{u\gamma^2}{2}\xi_\kappa(s)} ds = -\frac{2\kappa \bar{\nu}}{\gamma^2} \ln \left(1 - \frac{u\gamma^2}{2}\xi_\kappa(t) \right). \quad (18)$$

Collecting,

$$\mathbb{E}[e^{u\nu_T} | \nu_t] = \left(1 - \frac{u\gamma^2}{2}\xi_\kappa(\tau) \right)^{-2\kappa \bar{\nu}/\gamma^2} \exp \left(\frac{ue^{-\kappa \tau}}{1 - \frac{u\gamma^2}{2}\xi_\kappa(\tau)} \nu_t \right). \quad (19)$$

Although derived for real u , this extends to complex u within the convergence domain, and is exactly the cf of the (scaled) noncentral chi-squared variable above. This is the same Riccati mechanism that produces the full Heston cf below.

4.4 The Heston characteristic function

Now take the full log-Heston system, set $u = i\omega$ purely imaginary, and write $y_t = \ln S_t$. The transform $\mathbb{E}[e^{uy_T} \mid y_t, \nu_t] = e^{\alpha(\tau, u) + uy_t + \beta(\tau, u)\nu_t}$ has, with $\tau = T - t$ and the shorthands

$$b = \gamma\rho_{S,\nu}u - \kappa, \quad a = u(1 - u), \quad s = \sqrt{b^2 + a\gamma^2}, \quad (20)$$

the closed forms

$$\beta(\tau, u) = \frac{a(1 - e^{-s\tau})}{2s - (s + b)(1 - e^{-s\tau})}, \quad (21)$$

$$\alpha(\tau, u) = -r\tau + ru\tau - \kappa\bar{\nu} \left(\frac{s+b}{\gamma^2} \tau + \frac{2}{\gamma^2} \ln \left[1 - \frac{s+b}{2s} (1 - e^{-s\tau}) \right] \right). \quad (22)$$

The Heston characteristic function of the log-price is then

$$\boxed{\varphi_{\ln S_T}(\omega; \tau) = \mathbb{E}[e^{i\omega \ln S_T} \mid \ln S_t, \nu_t] = \exp(\alpha(\tau, i\omega) + i\omega \ln S_t + \beta(\tau, i\omega) \nu_t).} \quad (23)$$

This is the load-bearing object of the lecture. Everything — semi-analytic pricing, calibration, exact simulation of $\int \nu \, ds$ — runs through it.

Remark 2 (The “little Heston trap” / branch-cut subtlety). *The square root $s = \sqrt{b^2 + a\gamma^2}$ and the complex logarithm in (22) are multi-valued. The way the formula is grouped matters. The original Heston (1993) presentation uses an algebraically equivalent grouping (with a factor $g = (b + s)/(b - s)$ and $\ln(1 - ge^{s\tau})$ with $e^{+s\tau}$ rather than $e^{-s\tau}$). For long maturities τ that form repeatedly crosses the branch cut of the principal-branch $\ln$, the function becomes discontinuous and the integrand oscillates wrongly, giving wildly wrong prices. Albrecher et al. (2007) showed that the mathematically equivalent “little trap” formulation — precisely the one in (22), written with $e^{-s\tau}$ and $1 - \frac{s+b}{2s}(1 - e^{-s\tau})$ — stays on the principal branch and is numerically stable for all τ , provided one takes the principal square root with non-negative real part. The lesson: the two forms are equal as analytic functions but not as numerical recipes; always use the stable grouping.*

4.5 The pricing PDE (consistency check)

Itô on $e^{-rt}V(t, S, \nu)$ and setting the drift to zero (so the discounted price is a martingale) gives the Heston pricing PDE

$$\partial_t V + rS \partial_S V + \kappa(\bar{\nu} - \nu) \partial_\nu V + \frac{1}{2} \nu S^2 \partial_{SS} V + \frac{1}{2} \gamma^2 \nu \partial_{\nu\nu} V + \gamma \nu S \rho_{S,\nu} \partial_{S\nu} V - rV = 0, \quad (24)$$

where the cross term used $dS \, d\nu = \gamma \nu S \rho_{S,\nu} dt$. The cf (23) is the fundamental solution of the corresponding backward equation for the transform; this is the same PDE the Feynman–Kac/affine derivation invokes.

5 How the Heston cf plugs into COS

Recall from Lecture 06 the COS European pricing formula. With $x = \ln(S_0/K)$, density support truncated to $[a, b]$, and cosine coefficients V_k of the payoff,

$$v(x, t_0) \approx e^{-r\tau} \sum_{k=0}^{N-1} \text{Re} \left\{ \varphi\left(\frac{k\pi}{b-a}; x\right) e^{-ik\pi \frac{a}{b-a}} \right\} V_k, \quad (25)$$

where $\sum'$ halves the first term. The *only* model-specific input is the characteristic function $\varphi(\cdot)$ of the log-price. For Black–Scholes one substitutes the Gaussian cf; for Heston one substitutes exactly (23), evaluated at the COS frequencies $\omega_k = k\pi/(b - a)$:

$$\varphi\left(\frac{k\pi}{b-a}; x\right) = \exp\left(\alpha\left(\tau, \frac{ik\pi}{b-a}\right) + \frac{ik\pi}{b-a} x + \beta\left(\tau, \frac{ik\pi}{b-a}\right) \nu_0\right). \quad (26)$$

This is why the affine machinery matters for this course: COS is a quadrature of the inverse Fourier transform, and Heston is exactly the model whose density is unknown in closed form but whose cf is known in closed form, which is the regime where COS dominates.

Remark 3 (Truncation range under Heston). *The cumulant-based rule from Lecture 06, $[a, b] = [c_1 - L\sqrt{c_2 + \sqrt{c_4}}, c_1 + L\sqrt{c_2 + \sqrt{c_4}}]$, requires the cumulants c_1, c_2, c_4 of $\ln S_T$, which for Heston follow by differentiating $\log \varphi$ at $\omega = 0$. The danger I hit in Assignment 2 is generic: at short maturities τ the variance of $\ln S_T$ is small (small c_2), so $[a, b]$ becomes narrow; if the higher cumulants or a fat left tail (negative ρ) are under-weighted, the range can chop off mass and COS loses its spectral accuracy. Under Heston the skew from $\rho < 0$ and the vol-of-vol γ inflate the tails, so the prudent fix is a larger L (e.g. $L = 10\text{--}12$) and including c_4 , rather than trusting the BS-style $L = 8$ to 10 with c_4 ignored.*

6 Monte Carlo simulation under Heston

6.1 Euler scheme and the negative-variance problem

The naive Euler discretization of (5) on a grid t_i is

$$\nu_{t_{i+1}} = \nu_{t_i} + \kappa(\bar{\nu} - \nu_{t_i})(t_{i+1} - t_i) + \gamma\sqrt{\nu_{t_i}}\sqrt{t_{i+1} - t_i} Z, \quad (27)$$

$$\ln S_{t_{i+1}} = \ln S_{t_i} + \left(r - \frac{1}{2}\nu_{t_i}\right)(t_{i+1} - t_i) + \sqrt{\nu_{t_i}}\sqrt{t_{i+1} - t_i} Y, \quad (28)$$

with $Y = \rho_{S,\nu}Z + \sqrt{1 - \rho_{S,\nu}^2}X$, $X, Z \sim N(0, 1)$ independent. Because Z is Gaussian and unbounded below, $\mathbb{P}(\nu_{t_{i+1}} < 0) > 0$ always: the discretized variance can go negative, after which $\sqrt{\nu}$ is undefined. No naive time-stepping scheme avoids this.

Quick fixes (with errors).

- *Truncation / floor*: set $\nu_{t_{i+1}} \leftarrow \max(\nu_{t_{i+1}}, \varepsilon)$ for tiny $\varepsilon \geq 0$. Simple, but introduces a truncation bias.
- *Reflection*: set $\nu_{t_{i+1}} \leftarrow |\nu_{t_{i+1}}|$. Useful when the Feller condition fails; the reflected path systematically sits a little above the floored one near zero.
- *Log-variance change of variable*: simulate $Y_t = \ln \nu_t$ by Euler and exponentiate. This does *not* work: $\ln \nu$ can blow up, so $\nu = e^Y$ explodes for some paths.

These are all biased; the bias is worst exactly when the Feller condition is badly violated and the variance spends time near 0.

6.2 Exact simulation of the variance

A bias-free alternative for the variance uses the conditional law directly: $\nu_{t_{i+1}} | \nu_{t_i} \sim Y/(2c)$ with $Y \sim \chi_d^*(\lambda)$, $c = 2\kappa/[\gamma^2(1 - e^{-\kappa\Delta})]$, $d = 4\kappa\bar{\nu}/\gamma^2$, $\lambda = 2ce^{-\kappa\Delta}\nu_{t_i}$, $\Delta = t_{i+1} - t_i$. Sampling the noncentral chi-squared uses:

- if $d > 1$: $\chi_d^*(\lambda)^2 \stackrel{d}{=} (Z + \sqrt{\lambda})^2 + \chi_{d-1}^2$ with $Z \sim N(0, 1)$ and an independent ordinary χ_{d-1}^2 ;
- for general $d > 0$: draw $J \sim \text{Poisson}(\lambda/2)$, then conditionally $\chi_d^*(\lambda)^2 \sim \chi_{d+2J}^2$.

This is exact for ν and produces strictly non-negative samples. Caveat: the noncentral chi-squared representation needs $d > 0$, i.e. effectively the Feller condition; when it fails one switches to the log-variance transform of Fang–Oosterlee (2011).

6.3 Exact (Broadie–Kaya) simulation of the spot

Knowing the variance law is not enough for S , because the log-price depends on $\int_u^t \nu_s ds$ and on the stochastic integrals against W^ν, W^S . Integrating (1) over $[u, t]$,

$$S_t = S_u \exp\left(r(t-u) - \frac{1}{2} \int_u^t \nu_s ds + \rho_{S,\nu} \int_u^t \sqrt{\nu_s} dW_s^\nu + \sqrt{1 - \rho_{S,\nu}^2} \int_u^t \sqrt{\nu_s} dW_s^S\right), \quad (29)$$

$$\nu_t = \nu_u + \kappa \bar{\nu}(t-u) - \kappa \int_u^t \nu_s ds + \gamma \int_u^t \sqrt{\nu_s} dW_s^\nu. \quad (30)$$

The second line lets us express one stochastic integral in terms of the others. The Broadie–Kaya (2006) algorithm is then:

Step 1. Sample $\nu_t \mid \nu_u$ from the noncentral chi-squared above.

Step 2. Sample $\int_u^t \nu_s ds$ given ν_u, ν_t — *the hard step*.

Step 3. Recover $\int_u^t \sqrt{\nu_s} dW_s^\nu = \frac{1}{\gamma}(\nu_t - \nu_u - \kappa \bar{\nu}(t-u) + \kappa \int_u^t \nu_s ds)$.

Step 4. Sample $\int_u^t \sqrt{\nu_s} dW_s^S \mid \int_u^t \nu_s ds \sim N(0, \int_u^t \nu_s ds)$ (independence of W^S from the variance path).

Then assemble S_t . The result is unbiased and lets you jump directly from u to t with no intermediate time steps.

Step 2 via the cf of the integrated variance. The distribution of $\int_u^t \nu_s ds$ given (ν_u, ν_t) is obtained from its characteristic function $\varphi(\omega) = \mathbb{E}^\mathbb{Q}[\exp(i\omega \int_u^t \nu_s ds) \mid \nu_u, \nu_t]$, which is known in closed form (a ratio of modified Bessel functions $I_{d/2-1}$, with $R(\omega) = \sqrt{\kappa^2 - 2\gamma^2 i\omega}$ and $d = 4\kappa\bar{\nu}/\gamma^2$). One then inverts to get the CDF F — the original paper used the trapezoidal rule on the Fourier inversion; *today one uses the COS method* — and finally samples by inversion $\int \nu ds = F^{-1}(U)$ with $U \sim \text{Uniform}(0, 1)$, valid because $\mathbb{P}(F^{-1}(U) \leq r) = \mathbb{P}(U \leq F(r)) = F(r)$.

Trade-offs. Exact simulation has no discretization/truncation error, higher accuracy per path, and direct sampling at horizon t ; but it is slow and intricate to implement (Bessel functions, Fourier inversion per step). Broadie–Kaya’s tables show the exact scheme’s RMS error decaying as $O(N^{-1/2})$ in the number of trials with zero bias, versus Euler’s slower convergence dominated by discretization bias.

6.4 The three tools behind the integrated-variance cf (sketch)

The closed form for $\varphi(\omega)$ is built from three ingredients; knowing the names and roles is enough for the oral.

- **Infinitesimal generator.** For $dX_t = \mu(X_t) dt + \sigma(X_t) dW_t$ the generator is $\mathcal{A}f = \mu f' + \frac{1}{2}\sigma^2 f''$. The CIR generator is $\mathcal{A}f(x) = \kappa(\bar{\nu} - x)f'(x) + \frac{1}{2}\gamma^2 x f''(x)$. The n -dimensional *squared Bessel* process $dX_t = n dt + 2\sqrt{X_t} dW_t$ has generator $n f'(x) + 2x f''(x)$ and a known Laplace transform of its time integral (Pitman–Yor 1982).
- **Time change.** The scaling $cW_t \sim W_{c^2 t}$ lets us rescale the diffusion coefficient. The time change $s = \frac{\gamma^2}{4}t$ maps the CIR second-order coefficient $\frac{1}{2}\gamma^2 x$ onto the squared-Bessel value $2x$.
- **Change of measure (Girsanov).** After the time change the *drift* coefficients still differ ($\kappa(\bar{\nu} - x)$ vs n); a Girsanov change of measure absorbs the discrepancy, expressing the CIR integrated-variance Laplace transform as a ratio of squared-Bessel Laplace transforms, which Pitman–Yor give explicitly. Matching coefficients yields the Bessel-ratio formula for $\varphi(\omega)$.

7 Parameter intuition (for the smile)

- γ (vol-of-vol): with $\rho = 0$, larger γ deepens the smile curvature.
- $\rho_{S,\nu}$: more negative ρ steepens the skew (left tail heavier).
- κ : limited effect on the smile shape (1–2%); controls how fast ATM implied vol converges to $\sqrt{\bar{\nu}}$ in maturity.
- $\nu_0, \bar{\nu}$: similar effects, mediated by κ (ν_0 dominates short maturities, $\bar{\nu}$ long maturities).

Versus Black–Scholes, Heston gives heavier tails of log returns and reproduces the smile.

Honest gaps / things not on these slides

- The *full* explicit Riccati derivation of (21)–(22) (as opposed to the scalar CIR warm-up) is not carried out term by term on the slides; they quote the closed form. I have shown the mechanism on the 1-D CIR case, which is the same.
- The little-Heston-trap discussion (branch cut, Albrecher et al.) is *my addition*; the slides give the stable grouping (22) but do not flag the trap explicitly. Worth knowing for Fang’s exam.
- The exact integrated-variance cf (the Bessel-ratio formula, slide 53) and the full Girsanov derivation (slides 65–69) are stated; I have summarized them rather than reproduced every line. The squared-Bessel Laplace transform itself (Pitman–Yor) is quoted without proof.
- Cumulant formulas c_1, c_2, c_4 for Heston needed for the COS truncation range are referenced from Lecture 06 / Fang–Oosterlee; not derived here.